SMART INDIA HACKATHON — PROBLEM STATEMENT ID: 26261 AICTE REGULATION BUREAU • GOVT. OF INDIA

AI-Powered Application Validation, Document Authentication & Regulatory Compliance Verification

Client: All India Council for Technical Education (AICTE), Ministry of Education **Theme:** Smart Automation & DPI

Design Baseline: SIH26246 Liquid Glass Architecture

Core Statutory Mandate: Evidence-Linked Decision-Support Engine (Non-Black-Box)

This architecture operates strictly as an explainable, human-in-the-loop regulatory workstation for AICTE Scrutiny Committees (EVC/SHC) and Regional Officers. Every anomaly flag outputs pixel-level bounding boxes, Error Level Analysis (ELA) heatmaps, cryptographic verification hashes, and exact citations to the **AICTE Approval Process Handbook (APH) 2024–2027**. Final statutory approval or rejection authority remains exclusively with authorized officials.

1. Regulatory Context & National Scrutiny Bottlenecks

PART 1 • CONTEXT

AICTE processes over 8,500+ annual institutional dossiers across four statutory lifecycles: **(1) Extension of Approval (EoA)**, **(2) Increase/Variation in Intake & New Emerging Courses** (AI/ML, VLSI, Robotics), **(3) Establishment of New Institutions**, and **(4) Campus Closure or Site Relocation**. Each submission spans 150–400 pages of heterogeneous PDFs, notarized affidavits, CAD floor plans, and geotagged photographs. Manual verification suffers from five systemic vulnerabilities:

- **Severe Scrutiny Latency:** Manual cross-verification of Land Registry Deeds, Fire Safety NOCs, Building Completion Certificates, Barrier-Free (PWD) audits, University Affiliation letters, and Audited Balance Sheets creates multi-week bottlenecks during peak EoA windows.
- **Digital Document Forgery & Splicing:** Doctored certificate expiry dates, copy-pasted municipal seals, digitally synthesized architect signatures, and re-compressed JPEG patches hiding financial liabilities.
- **Cross-Institutional Fraud & "Ghost Faculty":** Faculty members simultaneously listed on the rosters of multiple private colleges to artificially satisfy the mandatory 1:15 / 1:20 Student-Faculty Ratio (SFR), alongside recycled laboratory equipment invoices and duplicate campus photographs.
- **Infrastructure & Blueprint Manipulation:** Inflated carpet areas in architectural drawings and falsified scale bars that mask undersized laboratories or missing circulation corridors.
- **Disconnected Issuing-Authority Verification:** Absence of automated cryptographic handshakes with DigiLocker/NAD, EPFO, MCA21, State Revenue Portals, and Municipal Fire Directorates.

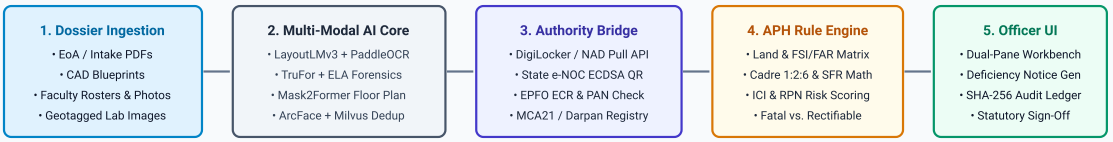


Figure 1.1: Five-Stage National Regulatory Scrutiny Pipeline integrating Multi-Modal CV, DPI Authority Handshakes, and Deterministic APH Rules.

DOSSIER THROUGHPUT	FORGERY LOCALIZATION	GHOST FACULTY RECALL
4.2 mins / App	**96.8% mIoU**	**99.4% @ 10^{-4} FAR**
Down from 14.5 hrs manual EVC review	Pixel-level splicing & date-alteration detection	512-D ArcFace + PAN/EPFO UAN triangulation

2. ML/AI Model Portfolio: IDP & Document Forgery Forensics

PART 3.1A–1B

To process heterogeneous Indian regulatory filings—ranging from digitally born financial spreadsheets to degraded 150-DPI scanner copies of vernacular land deeds—the platform deploys a hybrid cascade of state-of-the-art vision-language transformers and pixel-level forensic networks. Rather than relying on a single monolithic VLM prone to numeric hallucination, specialized backbones handle OCR transcription, spatial token classification, table topology reconstruction, and frequency-domain forgery localization.

Subsystem & Model	Architecture & Weights	Pre-Training & Loss Objectives	Fine-Tuning Hyperparameters	AICTE Target Artifacts
Text Detection & Recognition `PaddleOCRv4 + TrOCR`	DBNet++ (ResNet-50-vd) detector + SVTR-LCNet & TrOCR-Base printed/handwritten	Differentiable Binarization Loss ($L_s + \alpha L_b + \beta L_t$) + CTC & Cross-Entropy	AdamW, LR `1e-4`, cosine decay, batch 64, bilingual Eng + Devanagari tokens	Stamps, handwritten affidavit dates, vernacular land survey numbers
Layout & Key-Value Extraction `LayoutLMv3-Large`	24-layer Multimodal Transformer (1024-D, 16 heads, 16×16 patch linear projection)	Masked Visual-Language Modeling (MVLM) + Word-Patch Alignment (WPA) + BIO-CRF loss	AdamW (β_1=0.9, β_2=0.999, wd=0.01), LR `3e-5`, warmup 10%, seq len 512, batch 32	APH Format-1/2 Affidavits, Fire NOCs, PWD Certificates, Bank FDs
Table Topology & Financial Grid `Table-Transformer + Donut`	DETR Table Structure Recognition (TSR) + Swin-B OCR-free encoder-decoder	Bipartite Hungarian matching loss (L1 + GIoU) + Autoregressive token cross-entropy	AdamW, LR `5e-5`, input `1280×960`, 30 epochs with RandAugment scan noise	Multi-page Faculty Rosters (PAN, EPF, DOB), Audited Balance Sheets
Splicing & Tamper Localization `TruFor (CVPR 2023)`	Dual-branch RGB + Noiseprint++ Transformer (SegFormer MiT-B2 backbone)	Pixel-level Dice + Focal Loss (L_{loc}) + Confidence-weighted Integrity Score (L_{det})	AdamW, LR `6e-5`, poly schedule (power 0.9), crop `512×512`, JPEG Q∈[50,95]	Spliced NOC validity dates, copy-pasted municipal seals, altered land acres
Copy-Move & ELA Forensics `MantraNet + ResNet-ELA`	IML-ViT / Z-pool anomaly feature extractor + 90%-Q JPEG re-compression branch	Self-supervised 385-class camera trace contrastive loss + Binary Cross-Entropy	SGD w/ momentum 0.9, LR `1e-3`, dense SIFT + self-attention correlation matching	Cloned signatures across pages, duplicated lab invoice rows

3. Five-Layer Document Forgery & Metadata Forensic Pipeline

FORENSICS

- **Layer 1 — Error Level Analysis (ELA) Differential:** Re-compresses input page image I at 90% and 75% JPEG quality (I_{Q90}) and computes the amplified high-pass residual $E(x,y) = \gamma \cdot |I(x,y) - I_{Q90}(x,y)|$ (with gain $\gamma = 20$). Authentic regions exhibit uniform 8×8 DCT quantization decay, whereas spliced text patches exhibit high-variance boundary luminance spikes fed into a ResNet-50 anomaly classifier.
- **Layer 2 — Double JPEG Block Artifact Grid (BAG) Mismatch:** Extracts Discrete Cosine Transform (DCT) coefficient histograms across 8×8 macroblocks to detect localized phase shifts where a patch from another document was pasted at non-aligned grid coordinates $(\Delta x, \Delta y) \neq (0, 0) \bmod 8$.
- **Layer 3 — Copy-Move Forgery Detection (CMFD):** Combines dense RootSIFT keypoint extraction with a Vision Transformer self-correlation matrix $C_{ij} = (q_i \cdot k_j^T) / \sqrt{d}$ and RANSAC affine verification to flag cloned signatures or cloned numerical figures within the same multi-page PDF.
- **Layer 4 — Character Morphology & Baseline Kerning CNN:** Crops every recognized date, acreage number, and certificate serial string; computes character height variance, baseline tilt angle θ , stroke width consistency, and ink chromaticity in LAB color space to catch digits altered from `2021` to `2026`.
- **Layer 5 — PDF Cross-Reference (xref) & Ghostscript Forensics:** Inspects raw PDF byte streams via PyMuPDF (`fitz`) and `qpdf` for incremental updates (multiple `%%EOF` markers), non-standard `Producer`/`Creator` tags (`Adobe Photoshop`, `Canva`, `ilovepdf`), and subset font table collisions (e.g., a single line mixing `TimesNewRomanPSMT` with an unembedded `ArialMT` glyph).

Physical & Biometric Scrutiny: Bridging Paper Claims with Physical Reality

Beyond textual tampering, fraudulent institutions manipulate physical scale bars on architectural blueprints, submit stock photographs of engineering laboratories, and list "ghost faculty" employed at other colleges. Subsystems 1C, 1D, and 1E enforce cross-modal physical verification.

4. Signature & Institutional Seal Verification (Subsystem 1C)

PART 3.1C

Localization: A custom **YOLOv11m-Doc** detector (trained at `1024×1024` resolution with CIoU + Distribution Focal Loss) localizes four distinct classes on every page: `official_round_seal`, `registrar_stamp`, `handwritten_signature`, and `notary_emblem`. **Biometric Metric Learning:** Cropped regions undergo background text suppression via a CycleGAN clean-up module and are embedded using a **Siamese ResNet-50** backbone trained with **Additive Angular Margin Loss (ArcFace)**: $L_{Arc} = -log\ [\ e^{s \cdot cos(\theta_y + m)}\ /\ (e^{s \cdot cos(\theta_y + m)} + \sum_{j \neq y} e^{s \cdot cos\ \theta_j})\ ]$ with scale $s = 64$ and margin $m = 0.50\ rad$. Cosine similarity $S_{cos} = (e_{sub} \cdot e_{ref})\ /\ (\|e_{sub}\|\ \|e_{ref}\|)$ is evaluated against the institution's historical master signature/seal bank. Scores below `0.78` flag a mismatch; scores of `1.000` across distinct pages trigger an identical pixel-clone forgery alert.

5. Architectural Floor Plan & Area Compliance CV (Subsystem 1D)

PART 3.1D

Submitted CAD PDFs and scanned blueprints are rasterized at 300 DPI and segmented via **Mask2Former (Swin-Base backbone)** coupled with **YOLOv8x-seg** into structural primitives (`structural_wall`, `door_swing`, `window`, `staircase`, `fire_exit`) and functional zones. To prevent scale-bar manipulation, the pipeline executes dual scale calibration: (1) PaddleOCR extracts explicit dimension callouts (e.g., `"12.0m × 8.5m"`) and graphic scale bars, computing the pixel-to-meter ratio $\alpha = L_{meters}\ /\ L_{pixels}$; (2) structural doors are cross-checked against standard Indian National Building Code (NBC) widths (`0.9m–1.2m`). True room carpet area is computed via Green's theorem over the vectorized polygon vertices: $A_{m^2} = \alpha^2 \cdot ½\ |\sum (x_i y_{i+1} - x_{i+1} y_i)|$, validating directly against AICTE APH Chapter VII norms.

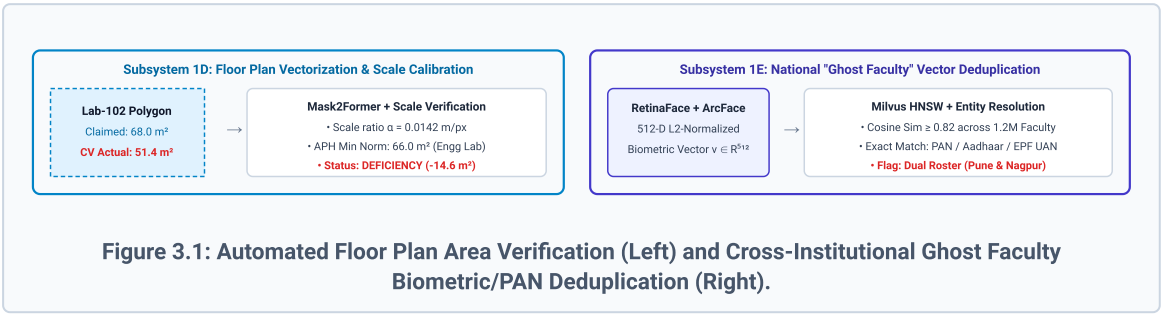


Figure 3.1: Automated Floor Plan Area Verification (Left) and Cross-Institutional Ghost Faculty Biometric/PAN Deduplication (Right).

6. Campus Scrutiny CV & "Ghost Faculty" Deduplication (Subsystem 1E)

PART 3.1E

- **Engineering Lab Equipment Verification:** Geotagged campus inspection photographs are analyzed by **RT-DETR-Large** fine-tuned on 42 AICTE laboratory asset classes (CNC lathes, Universal Testing Machines, Digital Storage Oscilloscopes, server racks, fume hoods, fire extinguishers). Furthermore, perceptual hashes (**PDQ + DINOv2 ViT-B/14 embeddings**) are queried against a national image ledger to detect colleges submitting identical photos of the same laboratory across different campuses or academic years.
- **EXIF GPS & Cadastral Polygon Triangulation:** Extracts raw EXIF tags (`GPSLatitude`, `GPSLongitude`, `GPSTimeStamp`, `Make/Model`) and verifies that the Haversine distance to the institution's registered AICTE master GIS centroid is within the surveyed land parcel polygon ($d \leq 250\ m$), flagging desktop screen recaptures via moiré frequency detection.
- **Ghost Faculty Biometric & Identity Graph:** Every faculty passport photo is aligned via **RetinaFace** and encoded via **ArcFace (ResNet-100 pretrained on Glint360k)** into a 512-dimensional vector indexed in **Milvus / Qdrant** using HNSW (`M=32, efConstruction=200`). A hybrid composite entity-resolution score fuses facial similarity ($S_{face} \geq 0.82$), deterministic hash collisions on `PAN_Number` and `EPFO_UAN`, and Jaro-Winkler fuzzy matching on `Name + DOB + Father_Name` to expose faculty double-counted across multiple institutions.

7. Exhaustive Datasets Catalog: Benchmarks & Domain Adaptation

PART 3.2A

Training high-precision forensic and document intelligence models requires a two-stage curriculum: Stage 1 pre-trains feature backbones on large-scale peer-reviewed academic benchmarks, and Stage 2 fine-tunes on our domain-specific Synthetic AICTE Regulatory Corpus to bridge the distribution gap to Indian stamp papers, bilingual layouts, and municipal records.

Task Domain	Public & Research Datasets	Scale / Volume	Ground-Truth Annotations	AICTE Pipeline Role
Document Layout & KIE	`DocBank`, `PubLayNet`, `FUNSD`, `CORD`, `SROIE`, `RVL-CDIP`	850K+ pages; 400K+ layout docs	Bounding boxes, reading-order graphs, BIO key-value semantic spans	Pre-training LayoutLMv3 & Table-Transformer on dense forms & receipts
Document & Image Tampering	`DocTamper` (CVPR'23), `CASIA v2.0`, `CoMoFoD`, `Coverage`, `NIST Nimble 16/17`	170K tampered text docs + 25K spliced images	Binary pixel-level manipulation masks (splicing, copy-move, removal)	Fine-tuning TruFor & MantraNet on character-level text splicing
Signatures & Official Seals	`CEDAR`, `GPDS-Synthetic`, `SigComp 2011/2013`, `Tobacco800`, `StaVer`	24K+ genuine/forged signatures; 12K stamps	Writer IDs, skilled forgery pairs, overlapping stamp segmentation masks	Training YOLOv11m-Doc detector & Siamese ArcFace verification head
Architectural Floor Plans	`CubiCasa5K`, `SESYD`, `CVC-FP`, `Rent3D`	5,200+ vector-annotated floor plans	Polygonal room masks, wall graphs, architectural symbol bounding boxes	Training Mask2Former & YOLOv8x-seg for blueprint vectorization
Biometric Face Deduplication	`Glint360k`, `MS1MV3`, `VGGFace2`, `LFW`, `IJBC`	17M images across 360K identities	Aligned 112×112 facial landmarks and identity class labels	Pre-training RetinaFace + ArcFace 512-D faculty embedding extractor
Bilingual Indian OCR	`BharatSceneText`, `IIIT-Indic-HW`, `Mozhi` (Hindi/Eng)	320K+ printed & handwritten word crops	UTF-8 Devanagari & Latin transcriptions with polygon quads	Fine-tuning PaddleOCR v4 & TrOCR on Indian notarized headers

8. Synthetic AICTE Dataset Factory (10,000+ PII-Safe Dossiers)

PART 3.2B

> **Privacy-Preserving Synthetic Generation Mandate (DPDP Act 2023 Compliant)**
> Because real institutional filings contain confidential faculty PANs, Aadhaar numbers, bank account details, and proprietary trust deeds protected under India's Digital Personal Data Protection (DPDP) Act 2023, we construct a deterministic + generative Python synthesis pipeline capable of producing **10,000+ labeled institutional dossiers** with exact pixel-level ground truth.

- **Recipe 1 — Synthetic Affidavits (APH Format-1 & Format-2) on e-Stamp Paper:** Jinja2 + WeasyPrint templates render authentic-looking Rs. 100 Non-Judicial e-Stamp headers (complete with synthetic 2D Aztec/QR barcodes, SHCIL serial numbers, and bilingual English/Hindi legal clauses) populated via a custom Indian-locale `Faker` provider generating valid-checksum synthetic PANs (`[A-Z]{5}[0-9]{4}[A-Z]{1}`), 12-digit EPFO UANs, and AICTE Permanent IDs (`1-1098452311`).
- **Recipe 2 — Controlled Forgery Injection Engine (Doctored Fire NOCs & PWD Certificates):** For 35% of generated certificates, an automated tampering script applies four realistic attack vectors while saving exact binary pixel masks $M(x,y) \in \{0,1\}$: (a) *Date Splicing*—replacing validity year `2023` with `2027` using a slightly mismatched font weight or 1.5° baseline skew; (b) *Patch Re-compression*—saving the pasted date patch at JPEG Q=65 inside a Q=92 host page; (c) *Copy-Move Stamp Cloning*—copying a notary seal from page 3 onto page 11 with a 4° rotation; and (d) *Metadata Injection*—re-saving via Ghostscript/Pillow with spoofed `Adobe Photoshop 25.0` XMP headers.
- **Recipe 3 — Procedural CAD Floor Plans with Intentional APH Non-Compliance:** Using `ezdxf` and `shapely`, we generate multi-wing institutional floor plans where 30% contain injected violations: e.g., drawing a Computer Lab

9. External Issuing-Authority Integration (Authority Bridge) PART 3.3

Forensic visual inspection alone cannot catch an authentic-looking document whose underlying approval was revoked by the issuing state department. The **Authority Bridge** microservice executes real-time, asynchronous cryptographic and REST handshakes across India's Digital Public Infrastructure (DPI) stack using mutual TLS (mTLS), OAuth 2.0 PKCE, and offline ECDSA public-key verification.

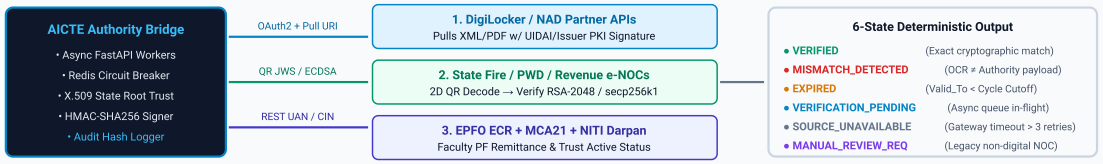


Figure 5.1: Multi-Agency Authority Bridge Architecture and Deterministic Six-State Verification Finite State Machine.

10. Verification Protocols, API Schemas & Fallback State Machine PROTOCOLS

- **DigiLocker & National Academic Depository (NAD) Pull Protocol:** Uses API Setu OAuth 2.0 with PKCE and HMAC-SHA256 request signing (`x-digilocker-hmac`). The engine queries `POST /pulldoc/v1/xml` using standardized URI patterns (e.g., `in.gov.maha.fire-NOC-2026-88412` or `in.ac.nad.degcert-...`). The returned W3C-compliant signed XML payload is verified against the Controller of Certifying Authorities (CCA) India Root PKI chain, and its canonical fields (`Certificate_No` , `Issued_To` , `Valid_Until` , `Survey_No`) are diffed against LayoutLMv3 extracted values from the uploaded PDF.
- **2D QR Code Cryptographic Verification (State e-District & Municipal Portals):** High-resolution QR/Aztec regions are cropped via YOLOv11, super-resolved via Real-ESRGAN if degraded, and decoded via `pyzbar` / `ZXing` . For JSON Web Signature (JWS) or CBOR Web Token (CWT) payloads, the engine validates the **ECDSA (secp256k1)** or **RSA-2048** signature offline against cached state public keys. If the QR contains a verification URL, a sandboxed headless worker fetches the official portal DOM and compares the live certificate record against the PDF text—instantly catching "Valid QR pasted onto a doctored PDF body."
- **EPFO ECR & MCA21 / NITI Darpan Handshake:** Cross-verifies the institution's submitted Faculty Roster against the Employees' Provident Fund Organisation (EPFO) Establishment TRRN (Temporary Return Reference Number) API to confirm active monthly PF remittances for every claimed full-time professor. Simultaneously queries MCA21 (Section 8 Company CIN) or NITI Aayog NGO Darpan (`Unique_ID`) to verify that the sponsoring Trust/Society registration is active and unencumbered.

> **Resilience, Circuit Breakers & Fallback Governance**
> State portals frequently experience intermittent downtime. The Authority Bridge wraps every external adapter in an exponential-backoff retry policy (`1s, 4s, 16s`) with a Redis circuit breaker. If a state endpoint fails after 3 attempts, the document transitions to `SOURCE_UNAVAILABLE` (never auto-rejected), while offline forensic checks (TruFor, ELA, QR signature verification) still execute immediately so scrutiny officers are never blocked.

11. AICTE Regulatory Rule Engine (APH Compliance Matrix)

PART 3.4

Extracted key-value entities, computer-vision floor plan areas, and Authority Bridge verification states converge into the deterministic + probabilistic **AICTE Approval Process Handbook (APH) Rule Engine**. Every stipulation is encoded as a versioned rule node linked to exact APH chapters and annexures.

APH Rule ID	Statutory Parameter	Mathematical Stipulation (APH 2024–27)	Verification Source	Violation Severity
`APH-CH1-1.4`	Contiguous Land & FSI / FAR	Mega/Metro: FSI norm (Min 1.5 acres) • Urban: **2.5 Acres** • Rural: **7.5 Acres** (Engg/Tech); max 2 parcels	Land Registry Deed + Cadastral GIS + Mutation	**FATAL_FRAUD** (if disputed/forged) / **MAJOR**
`APH-CH7-7.1`	Instructional Carpet Area	Classrooms ≥ **66 m²**; Tutorial ≥ **33 m²**; UG Engg Lab ≥ **66 m²**; Workshop ≥ **200 m²**; Drawing Hall ≥ **132 m²**	Mask2Former Blueprint CV + Architect Cert	**RECTIFIABLE** (if <10% short) / **MAJOR** (>10%)
`APH-CH7-7.2`	Admin, Amenities & Circulation	Principal Room ≥ 30 m²; Central Library ≥ **400 m²**; Boys/Girls Common ≥ 75 m²; Circulation = **0.25 × Carpet Area**	Mask2Former Blueprint CV + Geotagged Photos	**RECTIFIABLE** (14-day EVC compliance window)
`APH-CH7-7.3`	Student-Faculty Ratio (SFR)	UG Engg/Pharm: **SFR ≤ 1:20** • PG (M.Tech/MBA/MCA): **SFR ≤ 1:15**; Total Faculty $F_{req} = \lceil S_{sanctioned} / 20 \rceil$	Faculty Roster Table + EPFO ECR + ArcFace Dedup	**MAJOR** (Intake reduction penalty if SFR > 1:25)
`APH-CH7-7.4`	Faculty Cadre Ratio	**1 : 2 : 6** (Professor : Associate Professor : Assistant Professor); $N_{Prof} = F_{req}/9$, $N_{Assoc} = 2F_{req}/9$	Qualification Certs + Ph.D. NAD Check + PAN	**RECTIFIABLE** (Cadre shortfall notice)
`APH-ANX-16`	Mandatory Safety & Statutory NOCs	Valid Fire Safety NOC + Building Structural Stability + Barrier-Free PWD Ramp/Lift + Corpus Fund FDR	Authority Bridge QR/API + TruFor Forensics	**FATAL_FRAUD** (if doctored) / **MAJOR** (if expired)

12. Mathematical Scoring: ICI & Risk Priority Number (RPN)

ALGORITHMS

Institutional Compliance Index ($ICI \in [0, 100]$) & Fatal Fraud Veto Gate

To prevent a high infrastructural score from masking a forged Fire Safety NOC or cloned Land Deed, the engine computes a weighted compliance summation multiplied by a non-compensatory binary fraud veto gate $I_{fraud} \in \{0, 1\}$:

$$ICI = (1 - I_{fraud}) \times \sum_{k=1}^{5} w_k \cdot C_k - \sum_{m \in M} \lambda_m \cdot \Delta_m$$

where category weights are $w_{Land} = 0.20$, $w_{Infra} = 0.25$, $w_{Faculty} = 0.25$, $w_{Safety/NOC} = 0.20$, and $w_{Financial} = 0.10$; Δ_m represents percentage shortfall on rectifiable rules; and $I_{fraud} = 1$ if any mandatory document exhibits confirmed forgery ($P_{forge} \geq 0.85$) or deliberate cross-institutional asset duplication.

- **Risk Priority Number ($RPN \in [1, 1000]$) for Deficiency Triage:** Every flagged anomaly i is scored via Failure Mode and Effects Analysis (FMEA): $RPN_i = S_i \times O_i \times D_i$, where **Severity** $S_i \in [1, 10]$ measures statutory & life-safety impact (10 = forged Fire NOC or structural stability cert; 7 = ghost faculty inflating SFR; 3 = missing affidavit page stamp), **Occurrence/Confidence** $O_i \in [1, 10]$ scales with AI forensic probability ($\lceil 10 \cdot P_{anomaly} \rceil$), and **Detection Difficulty** $D_i \in [1, 10]$ weights stealth manipulations (e.g., 9 = subtle font kerning date splice or cross-state duplicate faculty; 2 = blank missing field).
- **Automated Decision Triage Bands:** `ICI ≥ 90 & Max RPN < 120` → **Fast-Track EoA Recommendation**; `65 ≤ ICI < 90 & $I_{fraud}$ = 0` → **Auto-Generate Structured Deficiency Notice (14-Day Rectification Window)**; `ICI < 65 or $I_{fraud}$ = 1 or Max RPN ≥ 500` → **Escalate to Standing Hearing Committee (SHC) & Physical EVC**

PART 4: MASTER PROTOTYPE BUILD PROMPT FOR ANTIGRAVITY (SECTION A — UI/UX & STACK)

Copy and execute Parts 4A and 4B directly inside the Antigravity autonomous agent workspace. The build must enforce 100% visual parity with the **SIH26246 "Liquid Glass Civic & Regulatory Intelligence UI"** specification and zero runtime mocks failure.

13. SIH26246 Design System & CSS Token Specification

BUILD PROMPT • UI

- **Aesthetic Formula ("Liquid Glass Civic Workstation"):** Reject dark cyberpunk clichés. Construct an authoritative Apple Vision Pro × Indian DPI workstation. Light canvas uses `#F4F7FB` with radial atmospheric mesh `radial-gradient(ellipse 80% 50% at 50% -10%, rgba(14, 165, 233, 0.06), transparent 70%)`; Dark mode uses `#0B132B` with subtle indigo/cyan depth.
- **Glass Surface Primitives (`globals.css`):** Define `.liquid-glass-card` with `background: rgba(255, 255, 255, 0.75)` (dark: `rgba(15, 23, 42, 0.75)`), `backdrop-filter: blur(24px) saturate(150%)`, `border: 1px solid rgba(226, 232, 240, 0.8)` (dark: `rgba(30, 41, 59, 0.8)`), and inner specular highlight `box-shadow: inset 0 1px 0 0 rgba(255, 255, 255, 0.95)`.
- **Typography & Bilingual Localization (EN | हिन्दी):** Load `Instrument Serif` (for editorial headers with `.title-serif-italic` accents), `Plus Jakarta Sans` (UI & body), `JetBrains Mono` (application IDs, SHA-256 hashes, APH rule codes, `tabular-nums`), and `Noto Sans Devanagari` + `Khand` for instant 100% English/Hindi dictionary switching via React Context.

14. Four Mandatory SIH26246 Component Patterns

COMPONENTS

1. Unified 6-Column KPI Grid

Container: `grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-3 xl:grid-cols-6 divide-y sm:divide-y-0 sm:divide-x divide-slate-200/80 dark:divide-slate-800 liquid-glass-card rounded-2xl overflow-hidden` . Each cell has a **2px top severity border** (`#DC2626` severe, `#D97706` moderate, `#059669` compliant, `#4338CA` archive), uppercase tracking label, and a **locked baseline row** aligning a `clamp(1.75rem, 2.4vw, 2.125rem)` tabular numeral with a delta pill badge.

2. Priority Intervention Alert Banner

Interactive container (`glass-interactive border border-amber-500/30 rounded-2xl p-4 sm:p-5`) with dual-ring pulsing indicator (`animate-ping` + solid core dot). Surfaces critical alerts (e.g., *"Apex Inst. of Tech — Land Deed Acreage Splicing Detected (98.4% Conf.)"*) with direct action triggers: `Inspect Forensic Heatmap` and `Issue SHC Show-Cause Notice` .

3. Multi-Layer App Shell & Filter Bar

Compact `240px` Left Glass Navigation Rail with AICTE Emblem, grouped routes, and active liquid-glass pill indicator. Floating Top Header with breadcrumb, live `Authority Bridge: 99.2% Online` indicator, ⌘K Command Palette, `EN | हिन्दी` toggle, and Global Contextual Filter Bar (Academic Year `2026-27` , Region, Inst Type, Category, Status).

4. Dual-Pane Scrutiny Workbench

Split-screen workspace in `/applications/[id]` : Left pane renders multi-page institutional PDFs on an HTML5 Canvas with toggleable layers (OCR Bounding Boxes, TruFor/ELA Heatmap Overlay, Stamp/Signature ROI, Floor Plan Polygons). Right pane displays extracted key-values, APH rule evaluation badges, and Authority Bridge live payload diffs.

15. Full-Stack Technical Architecture

STACK

Frontend: `Next.js 16 (App Router)` , `React 19` , `TypeScript 5.8` , `Tailwind CSS v4` , `Lucide React` icons, `Sonner` toast notifications, `next-themes` , and custom SVG/Canvas forensic overlays. **Backend Service (`/backend`):** `Python 3.12 + FastAPI` serving deterministic & ML endpoints (`/api/v1/scrutiny/analyze` , `/api/v1/forensics/ela` , `/api/v1/floorplan/segment` , `/api/v1/authority/verify` , `/api/v1/faculty/dedup`) powered by `PyMuPDF` , `OpenCV` , `NumPy/SciPy` , `ONNX Runtime` wrappers, and an embedded `DuckDB / SQLite` analytical store pre-seeded with realistic AICTE applications.

16. Complete 8-Route Architecture & FastAPI Endpoint Matrix

BUILD PROMPT • ROUTES

Next.js Route	Primary UI Components & Visuals	Interactive Capabilities & State	FastAPI Backend Binding
`/overview`	6-Col KPI Grid, Priority Alert Banner, India Regional Scrutiny SVG Map, Filing Feed	Filter by AICTE Region (North/West/South/East), click state to drill down, trigger batch scrutiny run	`GET /api/v1/dashboard/summary`
`/applications`	Searchable Institutional Registry Table, Category Pills (EoA, Intake, New), ICI Score Bars	Multi-facet filtering, sort by RPN Risk Score, instant search by AICTE PID / AISHE Code	`GET /api/v1/applications`
`/applications/[id]`	Dual-Pane Scrutiny Workbench (Interactive Canvas + APH Checklist & Field Inspector)	Toggle ELA Heatmap / OCR boxes / Stamp ROI; override AI flag with officer note; approve/flag	`GET /api/v1/applications/{id}` `POST /api/v1/scrutiny/override`
`/forensics`	Digital Forensics Studio: ELA Quality Slider, PDF xref Inspector, Siamese Stamp Comparator	Live upload or select dossier page; adjust JPEG Q (50–95%) & gain γ; view Cosine Sim matrix	`POST /api/v1/forensics/analyze`
`/infrastructure`	CV Floor Plan & Campus Audit Studio: Interactive Polygon Blueprint + Geotagged Photo Map	Click room polygon to inspect CV area vs. APH norm; verify EXIF GPS pins against land boundary	`POST /api/v1/cv/floorplan` `GET /api/v1/cv/campus-audit`
`/authority-bridge`	Live DPI Gateway Monitor (DigiLocker, State Fire Portals, EPFO ECR, MCA21) + Payload Diff	Simulate live Pull URI handshake, inspect ECDSA QR signature verification & retry queue	`POST /api/v1/authority/verify`
`/deficiencies`	Structured Deficiency Notice Generator + 14-Day Institutional Rectification Tracker	Auto-compile APH citations & forensic evidence into bilingual PDF notice; track EVC appeal	`POST /api/v1/deficiencies/generate`
`/audit-trail`	Immutable Scrutiny Ledger with SHA-256 Evidence Chain & Officer Digital Signatures	Verify cryptographic hash chain integrity; filter by Regional Officer ID, timestamp, action type	`GET /api/v1/audit/ledger`

17. Canonical Data Contracts & Pre-Seeded Demonstration Scenarios

SCHEMAS

Core TypeScript / Pydantic Schema Entities (`src/types/regulatory.ts`)

Define strict interfaces: `InstitutionalApplication` (`id, aictePid, aisheCode, name, state, region, category: 'EOA'|'INTAKE_INCREASE'|'NEW_INST'|'CLOSURE', iciScore, maxRpn, fatalFraudFlag, status, documents[], facultySummary, floorPlanAudit`); `DocumentArtifact` (`docId, type, pages, elaHeatmapUrl, truForScore, xrefAnomaly, stampSimilarity, authorityStatus: 'VERIFIED'|'MISMATCH_DETECTED'|'EXPIRED'|'VERIFICATION_PENDING'|'SOURCE_UNAVAILABLE'|'MANUAL_REVIEW_REQUIRED', extractedFields[]`); and `APHRuleEvaluation` (`ruleCode, chapter, parameter, requiredValue, actualValue, status: 'COMPLIANT'|'RECTIFIABLE_DEFICIENCY'|'FATAL_FRAUD', rpnScore, citation`).